

# Optimizing Squat Technique—Revisited

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## ABSTRACT

THE BACK SQUAT IS THE CORNERSTONE OF MANY STRENGTH AND CONDITIONING PROGRAMS, WITH INCREASES IN BACK SQUAT STRENGTH ASSOCIATED WITH IMPROVEMENTS IN ATHLETIC PERFORMANCE. A VARIETY OF BACK SQUAT TECHNIQUES HAVE BEEN DESCRIBED THROUGHOUT THE LITERATURE AND CAN BE OBSERVED ON SOCIAL MEDIA AND IN STRENGTH AND CONDITIONING FACILITIES, WITH NO CURRENT CONSENSUS REGARDING OPTIMAL TECHNIQUE. BASED ON CURRENTLY AVAILABLE EVIDENCE, A FULL DEPTH SQUAT, WITH A NATURAL FOOT POSITION, APPROXIMATELY SHOULDER-WIDTH APART, WITH UNRESTRICTED ANTERIOR MOVEMENT OF THE KNEES, AN UPRIGHT TRUNK, WITH A FORWARD AND UPWARD GAZE IS RECOMMENDED. THESE RECOMMENDATIONS SHOULD PERMIT INDIVIDUALIZATION BASED ON INDIVIDUAL ANTHROPOMETRICS. FOR A VIDEO ABSTRACT OF THIS ARTICLE, SEE SUPPLEMENTAL DIGITAL CONTENT 1 (SEE VIDEO, [HTTP://LINKS.LWWW.COM/SCJ/A241](http://links.lww.com/scj/A241)).

## INTRODUCTION

The back squat is one of the most popular and important exercises in the development of lower-

body strength and is regularly incorporated into strength and conditioning and rehabilitation programs (2,26). This is not surprising because maximal back squat performance shows strong correlations with performance in a variety of athletic tasks (16,20,66,73); with numerous studies demonstrating that short sprint performance improves with an increase in squat strength (15,17,56,57,61,65). However, athletes performing and coaches instructing the squat use a variety of techniques (e.g., foot position, stance width, depth) that are believed to reduce the risk of injury or be more beneficial for development of specific muscles or strength qualities (28), with no consensus regarding optimal technical performance of the back squat for over a decade (18). The purpose of this article, therefore, is to identify optimal back squat technique based on the evidence available from existing research. For the purpose of this article, we define “optimal” as the technique that not only minimizes injury risk, but also maximizes muscle activation of the prime movers and results in greater transfer to performance in athletic tasks. These recommendations should be appropriate for the majority of populations, although, possibly with the exception of powerlifters who tend to select a very wide stance to minimize displacement of the barbell.

## FORCES EXERTED ON THE KNEE LIGAMENTS

Contrary to popular belief, researchers have demonstrated that the back squat

does not place excessive strain on the anterior cruciate ligament (ACL) (9,26,27,39,50,52), even though compressive and shear forces to the knee increase as knee flexion increases (26). In fact, the back squat results in significantly less anterior displacement of the tibia, and therefore less stress on the ACL, when compared to open kinetic chain (OKC) exercises (e.g., leg extensions) (9,27,39).

Similar observations have been reported regarding posterior cruciate ligament (PCL) forces, with forces >4.5 times body mass during isokinetic and isometric leg extensions, compared with 3.5 times body mass during the back squat (27,68). These results indicate that the back squat results in lower strain and potential risk of strain on the PCL compared to leg extensions. However, it is worth noting that although PCL forces are lower in the back squat compared to leg extension, they progressively increase with increased knee flexion (64,75) and therefore with increased squat depth.

During the back squat, the ACL is only subjected to relatively low forces when the knee is <50° flexed, and at greater angles, the PCL rather than the ACL receives increased loads (28,68). Additionally, ACL forces were also shown to be much lower when the squat was performed with heels on the ground

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compared with when the heels were elevated during both the descent ( $\sim 365\%$ ) and the ascent ( $\sim 16\%$ ) (68).

Interestingly, when the back squat was included as part of off-season training, for professional soccer players, no increases in anterior-posterior tibiofemoral translation were observed at the end of the program (52). Therefore, it seems that not only is the back squat a potentially safer leg strengthening exercise than OKC exercises (9,27,39), but that when used consistently in training, the back squat does not increase ACL strain as identified by changes in tibiofemoral translation (52). Furthermore, preseason and in-season training focusing on increasing squat strength has demonstrated that not only does squat strength increase, but short sprint performance also improves (17,65).

A deeper squat tends to lead to increased anterior displacement of the knees, which may increase strain on the knee ligaments and menisci (4). Researchers have, however, demonstrated that restricting anterior displacement of the knees (so that the knees do not pass beyond the toes) results in a slight decrease in knee joint torque ( $\sim 22\%$ ); whereas restricting anterior movement resulted in a dramatic increase in hip joint torque ( $>1,000\%$ ) (29). The restricted squat (knees not passing beyond the line of the toes) also increases forward lean of the trunk (29,48), which was shown to noticeably increase lumbar shear forces (29). Therefore, restricting anterior displacement of the knee in an attempt to slightly reduce the forces exerted on the knee will dramatically and disproportionately increase force exerted on the hips and lumbar spine and is therefore not recommended (Figure 1).

### TRUNK POSITION

Dorsiflexion range of motion, assessed while the knee is flexed, has also been shown to be a predictor of an individual's squat depth (40) and is also negatively associated with forward lean of the trunk, whereby increased dorsiflexion results in a reduced forward lean of the trunk (30) (Figure 1). In



**Figure 1.** Unrestricted dorsiflexion (increased anterior displacement of the knees) versus restricted dorsiflexion (limited anterior displacement of the knees).

addition, static stretching of the gastrocnemius and soleus muscles, between sets of squats, has been shown to increase dorsiflexion and decrease forward lean of the trunk (19). Importantly, when static stretching, of the prime movers, was performed before performing repetitions to fatigue, using 80% of one repetition maximum (1RM), only a small non-significant decrease in the number of repetitions performed was observed (35). Therefore, for athletes with limited dorsiflexion and clear excessive forward lean of the trunk, it may be advantageous to perform static stretching of the lower leg musculature between sets of squats, although this may be best suited to the general preparation phase of a periodized training program, to avoid any small, acute, changes in force production, during the strength phase.

Additionally, the use of weightlifting shoes has been shown to increase the resultant level of dorsiflexion because of the heel lift allowing athletes to commence the exercise with plantar flexed ankles (46,67,72), which may aid in the reduction of forward lean of the trunk (46,67). For information on heel height (e.g., use of weightlifting shoes or heel wedge) and bar position (high versus low bar), see the review by Todoroff (67).

Interestingly, head position affects trunk position, with a downward gaze resulting in increased hip and trunk flexion and a forward or upward gaze resulting in a more upright trunk (23). It is therefore recommended that athletes maintain a forward or upward gaze to reduce lumbar shear forces which are a result of increased forward lean of the trunk.

## **IMPACT OF SQUAT DEPTH ON THE PATELLO-FEMORAL, TIBIO-FEMORAL AND HIP JOINTS**

Patello-femoral joint (PFJ) forces have been reported during the ascent phase of a deep squat when both rapid and slow ascents were used ( $4.73$  and  $5.99 \times$  body mass, respectively). PFJ forces were also similar during the descent phase of a deep squat using a rapid and slow ascent ( $7.62$  and  $7.41 \times$  body mass, respectively) (22). However, the results of another study demonstrated that PFJ forces increase, to a maximum of  $34.4$  N/kg, with knee flexion; however, knee flexion did not exceed  $90^\circ$  (70), whereas results of a further study demonstrated that PFJ forces increase with knee flexion during the squat and leg press (28).

Similarly, researchers have also reported little difference between tibio-femoral joint (TFJ) forces during fast ascent compared with a slow ascent ( $5.01$  and  $4.72 \times$  body mass, respectively) and during a fast descent compared with a slow descent ( $7.62$  and  $7.41 \times$  body mass, respectively) (70). Speed of movement does not dramatically alter TFJ forces during deep squatting although TFJ forces progressively increase with increased knee flexion (28,33).

Interestingly, the quadriceps tendon may not be subjected to notably greater loads during the back squat compared to a paused back squat (1 second pause at the bottom of the descent), although quadriceps tendon stress and strain was greater during the squat jump performed at comparable loads to the back squat (24). It is important to note, however, that higher movement velocities during the squat result in higher ground reaction forces (8), and that the intention to move quickly rather than the actual velocity of movement has been shown to determine the adaptive responses in relation to rapid force production (7).

It is also worth noting that although lighter loads are used when squats are performed to a greater depth, this decrease in load does not offset the

increased loading on the knee joint because of the greater angles resulting in increased joint moments (21). Therefore, individuals with PFJ or TFJ pain may be better suited to performing the back squat to a reduced, pain free range of motion specific to their sport. It is important, however, that athletes who regularly incorporate the clean and snatch into their training are appropriately conditioned to squat below  $90^\circ$  because of the “catch” or “receive” phase of both lifts as knee flexion can reach  $\sim 118^\circ$  (63). The reduced barbell loads associated with a deeper squat are likely to result in decreased spinal loads, with it being suggested that squat depth does not increase injury risk (59) if good technique is adopted and maintained.

In individuals with femoral acetabular impingement (FAI), pain is commonly perceived during a deep squat, potentially limiting range of motion (42–45). Therefore, in individuals with FAI, it has been suggested that squat depth should be maintained within a comfortable, pain-free, range of motion with the legs abducted and feet externally rotated (5,42,43).

## **IMPACT OF SQUAT DEPTH ON PERFORMANCE**

Deeper squats have also been reported to result in greater transfer to performance in jump performance and both deep and shallow 1RM squat performance, than shallow squat performance, albeit that this study used machine rather than free weight squats (71). Moreover, deep front squats and back squats have been reported to result in improvements in 1RM squat performance and squat jump and countermovement jump performances, whereas no improvements were reported in subjects who performed quarter ( $60^\circ$  knee flexion) back squats, when using the same loads (34). Such findings should not discourage the use of partial range of motion (shallow) squats because the result of a more recent study demonstrated the addition of shallow squats to full depth squats results in greater improvements in

1RM and impulse (0–50-, 0–90-, 0–200- and 0–250 ms) during an isometric squat (6). It should be stressed that these results were the addition of shallow squats to deep squats and not the replacement of deep squats. In addition, the authors hypothesized that the improvements were due to the increased intensity permitted during the shallow squats.

Because increasing squat depth results in a decrease in the load that can be lifted, Rhea et al. (55) compared the effects of squat depth ( $1/4$ ,  $1/2$  and full depth) using loads relative to the 1RM in the squat depth trained (resulting in the highest loads in the  $1/4$  squat), in well-trained collegiate athletes. The results demonstrated the greatest increase in 1RM at the depth that was trained, but greater improvements in vertical jump and short sprint performance in the  $1/4$  squat group.

## **MUSCLE ACTIVATION**

There is debate among strength and conditioning practitioners regarding the most appropriate foot placement and squat depth, not only in terms of stresses on the knee and hip joints and the spine, but also in terms of muscle activation. Importantly, the findings of a recent study highlighted that muscle activation, in all muscles studied, was higher in a free weight back squat compared to the Smith machine back squat, although this was only significantly higher for the gastrocnemius, the vastus lateralis, and the biceps femoris (60).

## **IMPACT OF SQUAT DEPTH ON MUSCLE ACTIVATION**

The results of electromyographic (EMG) studies have clearly demonstrated that increased squat depth results in a greater percentage contribution from the gluteus maximus (14,51). Vastus medialis data suggest a trend in which the contribution of the vastus medialis increases with increased squat depth (14). More recently, Bryanton et al. (13) reported that knee extensor activation increases with squat depth, although





**Figure 2.** Correct lower leg alignment, with knees tracking over toes versus incorrect knee alignment with knee valgus.

hip extensor activity increases with both squat depth and load lifted, quadriceps activation has been shown to depend on hip extensor muscle strategy (12). During the eccentric phase of the weighted-back squat, the relative contributions of 4 muscle groups (vastus medialis, vastus lateralis, biceps femoris, gluteus maximus) at the 3 depths tested were not statistically different for both average and peak EMG (14). Increased squat depth has, however, been shown to result in an increased vastus medialis and vastus lateralis activation ratio (38).

#### **IMPACT OF FEET ORIENTATION AND STANCE WIDTH ON MUSCLE ACTIVATION**

Rotating the feet (e.g., neutral, 30–40° medial, 80° lateral rotation) while performing the squat, regardless of

depth and stance width (75–140% shoulder width), has been shown to have no noticeable effect on leg muscle activity (rectus femoris, vastus medialis, vastus lateralis, adductor longus, semimembranosus, semitendinosus, and biceps femoris) (27,28,49,51,62). Greater external rotation of the feet should result in increased external rotation of the hip joint, however, this did not alter quadriceps muscle activity, but did increase hip adductor activity (54). Additionally, neutral, valgus, and varus positions of the hip do not seem to alter the ratio of activation of the vastus medialis and vastus lateralis; however, as previously mentioned, increased squat depth does result in increased vastus medialis activation and the vastus medialis and vastus lateralis activation ratio (38). External

rotation of the feet may also permit a greater squat depth.

Increased stance width may not alter muscle recruitment patterns of the quadriceps, but does lead to increased activity of the adductor longus (> shoulder width) and gluteal muscles (49,51,53) when a wide stance is used. It is worth noting that this increase in muscle activation may increase force production and therefore performance during the execution of the lift. It is therefore suggested that a natural, self-selected, foot width and placement, which permits a full range of motion, is adopted during the squat because this does not noticeably alter muscle recruitment of the primary musculature involved.

#### **HAMSTRING MUSCLE ACTIVATION DURING THE BACK SQUAT**

Contrary to popular belief, research findings indicate that the back squat, regardless of stance width, foot placement or squat depth, produces low levels of activity in hamstring muscles (3,4,27,28,32,37,49,51,52,58,60), with only poor technique including excessive forward lean likely resulting in high levels of muscle activation. To ensure balanced leg training, it would be prudent to include hamstring-specific exercises, such as the leg curl and stiff-leg (Romanian) deadlift because they showed greater hamstring activity than the squat (3,74). Ideally, hip and knee dominant exercises should be incorporated into the training regime to ensure optimal architectural development of the biceps femoris (10).

It is also important, therefore, that coaches do not assume that the hamstrings will be effectively trained during a back squat and therefore adequately plan and program hamstring strengthening via alternative means. This should ensure adequate conditioning of the hamstring muscles, in comparison with the quadriceps to ensure adequate force generation during cocontraction to minimize both anterior and lateral tibial translation (25,47), decrease shear

forces (41), and increase knee joint stability (1) both during the squat and a variety of athletic tasks. Increasing the force of cocontraction of the hamstrings reduces the forces exerted upon the ACL during knee flexion (47) and therefore may reduce risk of injury (36). Similarly, appropriate eccentric conditioning of the hamstrings has been shown to decrease the risk of hamstring strain injuries (11,31,69).

## CONCLUSION

Based on the literature reviewed, the optimal squat technique to minimize the risk of injury and ensure maximal activation of the lower limb muscles should be a natural stance width and foot positioning, with unrestricted movement of the knees (heels remain in contact with the floor) (Figure 1), gaze forwards or upwards, and full depth (115–125° knee flexion), as long as the spine is maintained in a neutral position. Appropriate alignment of the lower leg during the squat is essential, with the knees tracking over the toes and no knee valgus (Figure 2).

It has previously been recommended that squats should be performed in a more “functional” range of 0–50° of knee flexion (28,50,73) because forces on the PCL, PFJ, and TFJ increase with increased flexion (28,64). However, for any athletes using or planning to use the clean or snatch within their training, it is essential that they incorporate the full-depth squat into their training regime because it is a vital component of the “catch” phase of the clean and the snatch, where knee flexion can reach 118° (63). Additionally, greater improvements in performance seem to be achieved when squats are performed through a greater range of motion (34,71), or when shallow squats are used as a supplementary exercise to full depth squats (6).

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